

Statistics

z score : (universal translator)

— z score tells how far a point is from the mean

→ unit of z score : standard deviation

example

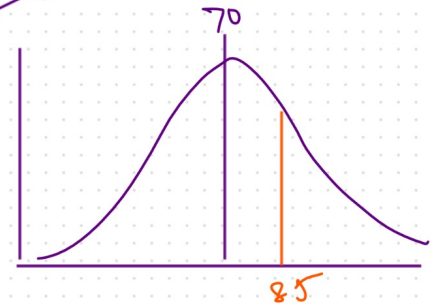
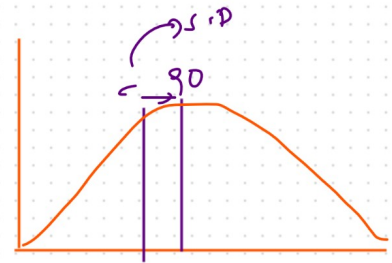
class A

scored 85 marks

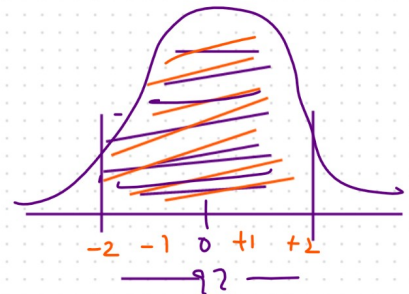
history
maths

maths → students → average 90
low

history → student — 70
— high



$$z_{score} = \frac{x - \mu(\text{mean})}{\sigma(\text{S.D.})} = \frac{x - \mu}{\sigma}$$



$$\oplus 2 \rightarrow 0.9772 \rightarrow 97\%$$

area under curve probability
→ z-table

P.S

height

$$\sigma = 10 \text{ cm}$$

$$\mu = 150 \text{ cm}$$

$$x = 170 \text{ cm}$$

$$\frac{170 - 150}{10} = \frac{20}{10} = 2$$

97% taller than your peers

Central Limit theorem :

{ . . . } Students (1000)

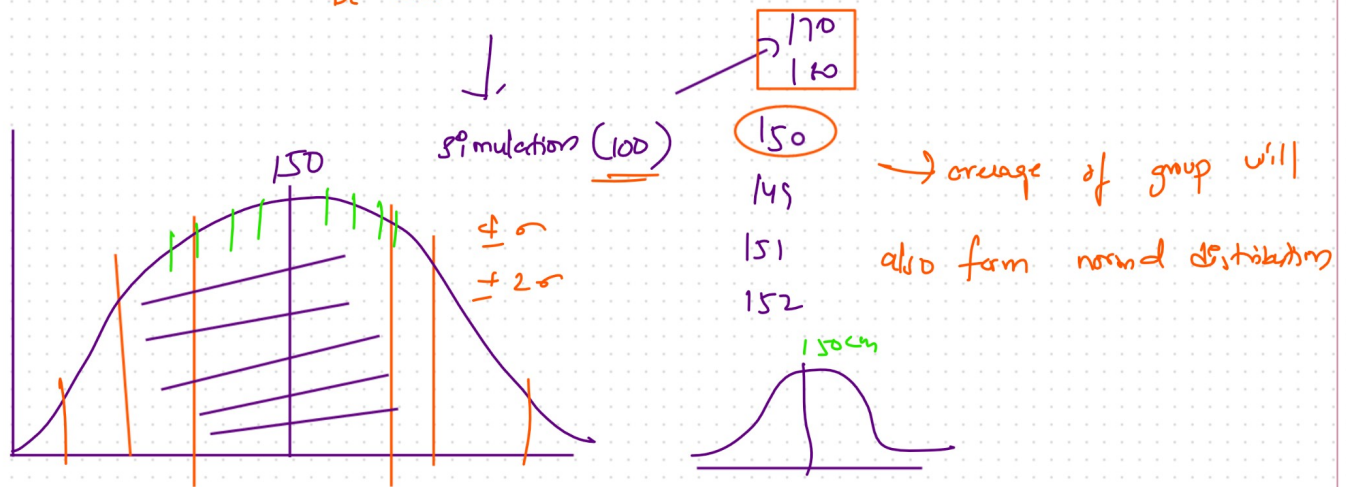
(μ) $\bar{x}$ — 150 cm
true population
 $\bar{x} \rightarrow 150 \text{ cm}$

extremely short (100cm)

extremely tall (200cm)

if you pick 100 random students — group
↓
take their average height (mean)

↓
that average height will almost always
be close to the true average



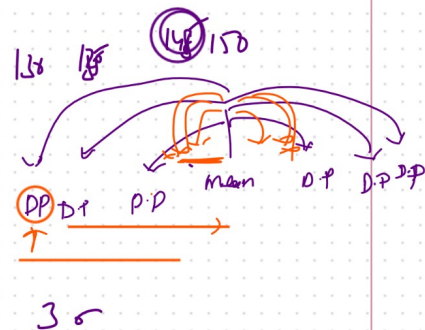
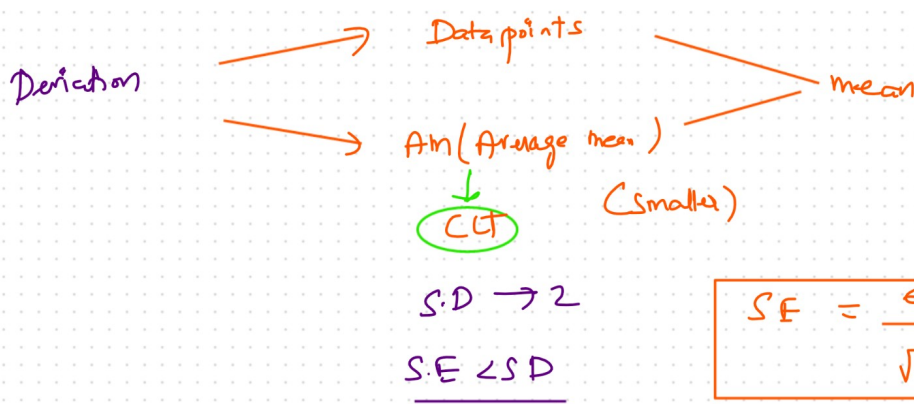
Conclusion: take sample size $n \geq 30$
↓
no. of data point

Standard Error (SE)

SD: measure the spread of individuals (datapoints) from the mean.

SE: measure the spread of group average from the mean

- ① mean $\rightarrow$ SD $\rightarrow$ which one is near (close to mean)
② mean $\rightarrow$ SE $\rightarrow$



$$SE = \frac{\sigma}{\sqrt{n}}$$

Scenario

Group of 25 students

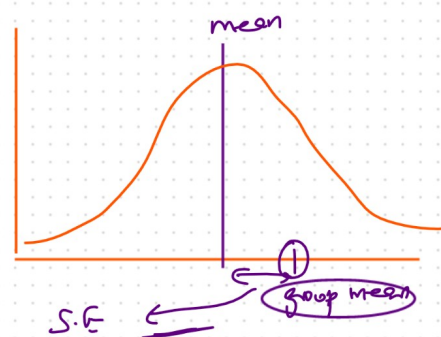
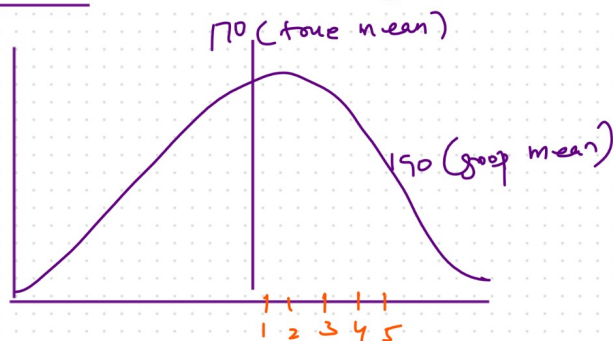
$$SD(\sigma) = 10 \text{ cm}$$

$$S.E = \frac{\sigma}{\sqrt{n}} = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2$$

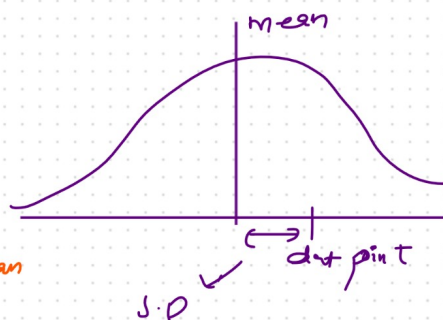
$$S.E < S.D$$

$$2 < 10 \quad \checkmark$$

Z-statistics : how many Standard Error (SE) fit into the gap



Z-score : how many SD fit into the gap



$$Z_{\text{stat}} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$\frac{\text{sample mean} - \text{true mean}}{S.E}$$

Scenario

The claim: A company says that their energy drink has 200 mg Caffeine on average

The evidence:

population mean (μ) = 200 mg

population deviation (SD or σ) = 10 mg

↓ ↓ ↓
190 200 210

sample size (n) = 40 cans

sample mean $\bar{x}$ = 198 mg →

Question : Is our 198 mg result "significantly different" or just lucky/unlucky random sample?

$$z_{\text{stat}} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} \rightarrow \text{diff}$$

$$198 - 200 = -2$$

$$\frac{10}{\sqrt{40}} \approx 1.581$$

$$\rightarrow \frac{-2}{1.581} \approx -1.265$$



95% ✓

198 is lower than 200, but is not weird enough to prove that company is lying. It's probably just a random coincidence